AI Outreach Agent

An agent that runs a targeted job search end to end, with a human approving everything it sends

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github.com/Beethoven04/ai-outreach-agent

The problem. I wanted a master's thesis internship at AI companies in Skåne and Copenhagen. Doing it well by hand means finding the right companies, the right person in each team, a real email address, a personal email for each one, follow-ups and answers, without ever sending a wrong fact or a duplicate. I built a system that does the work and keeps me in control of every decision that matters. **It has run in production since October 2026.**

HOW IT WORKS

1 · Research	2 · Verify	3 · Write + check	4 · I approve	5 · Send	6 · Replies
Job-ad APIs, startup and incubator lists, company team pages	Infer the address format, confirm the mailbox with the mail server	LLM drafts; code rules + LLM rubric check every claim	One tick in the sheet. Nothing leaves without it	Safe schedule, caps, gaps, duplicate checks in Gmail	Classified, referrals followed, answers drafted on Telegram

Nine n8n workflows and a Python research tool share one Google Sheet as the database, readable by a person and by code. A daily Telegram report shows what was sent, what came back and any problem with its fix.

METHODS

Find companies	JobTech open job-ad API (Sweden), The Hub API, incubator and science-park portfolios, Hunter Discover
Find the person	Team and career pages, job ads that name the hiring manager. Founders at startups, team leads at large firms
Find the address	Published > public GitHub commit address > company format inferred from real staff addresses
Confirm it	DNS MX → catch-all test (one random address per domain) → self-hosted Reacher SMTP check → Reoon for Microsoft 365
Write it	Gemini Flash, Groq fallback. Each email names one concrete reason that links the company to my work
Guard it	Code checks what can be checked (length, dates, names, banned phrases). The LLM must quote the text for every finding and match each claim to my profile facts

ACCURACY

<0.5%	68%	0	172
bounce rate on cold emails	of people found got a mailbox-confirmed address	guessed addresses sent automatically	automated end-to-end checks

The other 32% were held back (catch-all domain or no confirmation), not guessed. About half of the confirmed addresses came from an inferred company format, 39% were published. A second verifier settled 19%, mostly Microsoft 365 domains where a plain SMTP check gives a false "invalid".

ENGINEERING

- **Workflows as code:** every workflow is generated from JavaScript and Markdown prompts, diffable and tested before deploy.
- **Tested like software:** every code step runs against an anonymised sheet with all APIs mocked, then in a real n8n container.
- **Safe by default:** content hash after QC, append-only log, Gmail "Sent" check before each send, auto-pause on quota errors.

WHAT PRODUCTION TAUGHT ME

- **Verifiers can be wrong.** Microsoft 365 throttles the second recipient in one SMTP session, so real mailboxes looked invalid. Fix: a second verifier for those domains.
- **Quotas hit per minute, not per day.** A burst of Gmail reads beside a send tripped the limit. Fix: each Gmail workflow got its own minute of the hour.
- **People write by hand too.** An hourly Gmail-to-sheet sync logs my own emails, so the agent never sends on top of them.
- **Natural voice matters.** The QC now flags phrases that read as machine-written, and the style rules were rewritten around plain, short sentences.

Did you receive an email from me? Then you have seen this system at work: your company and your name came from the research step, your mail server confirmed the address, an LLM drafted the email, the checks matched every claim to my real profile, and I read and approved it before it was sent.

n8n · Python · JavaScript · Gemini · Groq · Google Sheets API · Gmail API · Telegram Bot API · Docker · Reacher · Claude Code (role-based agent workflow)

Public sources only. No scraped leak databases, no LinkedIn scraping. Mailbox checks never send mail.

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